

The Three Matrix Spaces

The Null Space of a Matrix

Defn. The **null space** of matrix A , denoted $Nul A$, is all solutions to the homogeneous system $Ax = 0$. That is, all vectors mapped to 0 by the matrix transform $x \mapsto Ax$.

If A is an $m \times n$ matrix, then $Nul A$ is a subspace of $\mathbb{R}^n$.

Basis for Null Space

ALGOR ***Basis for Null Space***

A basis for $Nul A$ is obtained by creating one vector for each free variable after row reduction. The dimension of the null space is the number of free variables.

The Column Space of a Matrix

Defn. The **column space** of matrix A , denoted $\text{Col } A$, is all linear combinations of columns of A .

If A is an $m \times n$ matrix, then $\text{Col } A$ is a subspace of $\mathbb{R}^m$.

Basis for Column Space

ALGOR ***Basis for Column Space***

A basis for $\text{Col } A$ is obtained by taking each ***original*** column vector that corresponds to a pivot column. The dimension of the column space is the number of basic variables.

The Row Space of a Matrix

Defn. The **row space** of matrix A , denoted $\text{Row } A$, is the set of linear combinations of rows of A .

If A is an $m \times n$ matrix, then $\text{Row } A$ is a subspace of $\mathbb{R}^n$.

Fact. If two matrices are row equivalent, then they have the same row space.

Basis for Row Space

ALGOR ***Basis for Row Space***

A basis for the row space is obtained by taking the nonzero rows of echelon form. The dimension of the row space is the number of pivots.

Example of The Three Bases

$$\begin{bmatrix} 1 & -1 & 4 & 7 & -1 \\ 5 & -5 & 0 & 15 & 11 \\ 2 & -2 & -2 & 4 & 6 \end{bmatrix} \text{ reduces to } \begin{bmatrix} 1 & -1 & 0 & 3 & 11/5 \\ 0 & 0 & 1 & 1 & -4/5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Null space: dim 3

$$\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -11/5 \\ 0 \\ 4/5 \\ 0 \\ 1 \end{bmatrix}$$

Col. space

dim 2

$$\begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix}$$

Row space

dime 2

$$(1, -1, 0, 3, 11/5) \\ (0, 0, 1, 1, -4/5)$$

The Rank of a Matrix

Fact. For any matrix, its row and column spaces have the same dimension.

Defn. The **rank** of a matrix A is the dimension of $\text{Col } A$ or equivalently the dimension of $\text{Row } A$.

Rank and Invertible Matrices

Fact. *Matrix A is invertible $\iff A$ has full rank.*

Summary

The null space of matrix A is all solutions to the homogeneous system $Ax = 0$. The column space of matrix A is all linear combinations of columns of A . The row space of matrix A is the set of linear combinations of rows of A .

Summary (cont)

A basis for the null space has one vector for each free variable after row reduction. A basis for the column space is obtained by taking each original column corresponding to a pivot column. A basis for the row space is obtained by taking the nonzero rows of echelon form.

Dimension of null space is the number of free variables. Dimension of column and row spaces is number of pivots; this is the rank of the matrix. Matrix is invertible if and only if full rank.